

ConDiff-Seg: Medical Image Segmentation Algorithm Based on Conditional Diffusion Model



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Abstract Medical image segmentation is essential for clinical diagnosis and treatment; however, existing methods still suffer from insufficient robustness in cross-modal and other complex tasks. In recent years, diffusion models have attracted attention for their stability and strong generative capabilities, and have been increasingly applied to segmentation tasks. Nevertheless, most studies are confined to single scenarios and lack systematic validation. This paper systematically evaluates several conditional diffusion-based segmentation methods and conducts comparative experiments on multiple public datasets. Experimental results show that DiffAtlas achieves the best performance on core metrics, IoU, Dice, and SEN, and, in particular, demonstrates stronger stability and generalization in few-shot and cross-domain tasks. EIDiffSeg attains performance on Dice and SEN close to that of DiffAtlas, while EnsemDiff stands out in PPV, although its overall robustness is slightly lower. These findings validate the effectiveness of diffusion models for medical image segmentation and highlight the advantages of a generative framework that jointly models images and masks in complex medical scenarios, providing valuable references for clinical applications and future research.

Keywords Diffusion models · Medical image segmentation · Cross-modal · Complex scenarios

1 Introduction

In recent years, neural network-driven generative modeling has rapidly emerged in the field of deep learning, demonstrating considerable potential across various tasks. The core idea of generative models is to simulate the data generation process and produce new samples from learned distributions, with applications spanning images,

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speech, text, and point clouds [1]. Early generative models include energy-based models (EBMs), generative adversarial networks (GANs), variational autoencoders (VAEs), and normalizing flows (NFs) [2]. While these approaches have improved the diversity and quality of generated results to some extent, they commonly suffer from unstable training, mode collapse, or inadequate posterior estimation. In contrast, diffusion models (DMs) [3] provide a novel paradigm for generative modeling, owing to their stable training objectives and strong distribution modeling capabilities [4]. The core mechanism involves gradually transforming complex distributions into Gaussian noise via a forward diffusion process, followed by stepwise reconstruction of sample data through a reverse diffusion process, thereby achieving superior visual fidelity and generation stability [5] (Fig. 1).

In the field of medical imaging, accurate image segmentation is critically important for clinical diagnosis and treatment decision-making [6]. Traditional approaches rely on handcrafted priors or deep models based on convolutional neural networks (CNNs), such as U-Net and its variants, which effectively capture both local and global features via encoder-decoder architectures [7]. However, these methods often exhibit limited robustness and unstable boundary delineation when faced with low-quality images, complex noise environments, or small lesions [8, 9]. More recently, the introduction of Vision Transformers (ViT) and their improved architectures [10] have further advanced the diversification and refinement of medical image segmentation methods; nevertheless, they may still encounter challenges related to inadequate feature modeling in complex scenarios.

The emergence of diffusion models has opened new avenues for medical image segmentation. Compared with GAN-based models, diffusion models exhibit higher distribution coverage and more stable optimization objectives, enabling high-quality segmentation performance even under noisy conditions [11]. Existing studies have demonstrated that diffusion model-based methods achieve superior performance across multiple tasks. For instance, the MedSegDiff model proposed by Wu et al. [12] achieves more accurate segmentation results through dynamic conditional encoding and multi-scale prior fusion; its subsequent version, MedSegDiff-V2 [13], further enhances model stability and accuracy by integrating a Transformer framework. In addition, Wolleb et al. [14] proposed a diffusion probabilistic model-based segmentation approach capable of generating diverse masks through multiple stochastic samplings, which, after implicit ensembling, leads to improved segmentation performance. Furthermore, the PD-DDPM method proposed by Guo et al. [15] effectively reduces the number of reverse diffusion steps and accelerates the segmentation process by combining pre-segmentation results with noise prediction. Collectively, these works highlight the potential and innovativeness of diffusion models in medical image segmentation tasks.

However, existing studies have predominantly focused on specific public datasets or segmentation tasks involving a single disease, and their performance on broader, more diverse medical datasets has yet to be systematically validated. To address this limitation, the present study applies various existing diffusion model-based methods to datasets different from those used in previous research and conducts

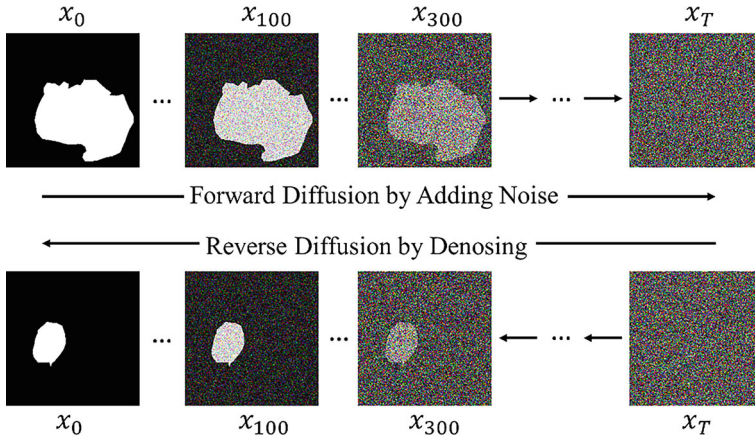


Fig. 1 Schematic illustration of diffusion models, which perturb data with noise and subsequently denoise to generate samples

cross-dataset comparative experiments to comprehensively evaluate their performance and applicability across different medical scenarios. Through this investigation, we aim to address the following key questions: which diffusion model-based segmentation method achieves the best performance on novel medical imaging datasets, and what are the respective strengths and limitations of these methods in fine-grained lesion identification and boundary delineation? This exploration not only contributes to advancing the clinical applicability of diffusion models in medical image segmentation but also provides new perspectives and references for future research.

2 Methods

2.1 Standard Diffusion Models Network

The standard diffusion skeleton network comprises two stages: a forward noise addition process and a reverse denoising process, as illustrated in Fig. 2. In the forward diffusion process, the model defines a fixed Markov chain that gradually injects noise into the segmentation mask until the data is entirely degraded into an approximate standard Gaussian distribution. Each transition step is described by:

$$q(x_t|x_{t-1}) = N(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t I) \quad (1)$$

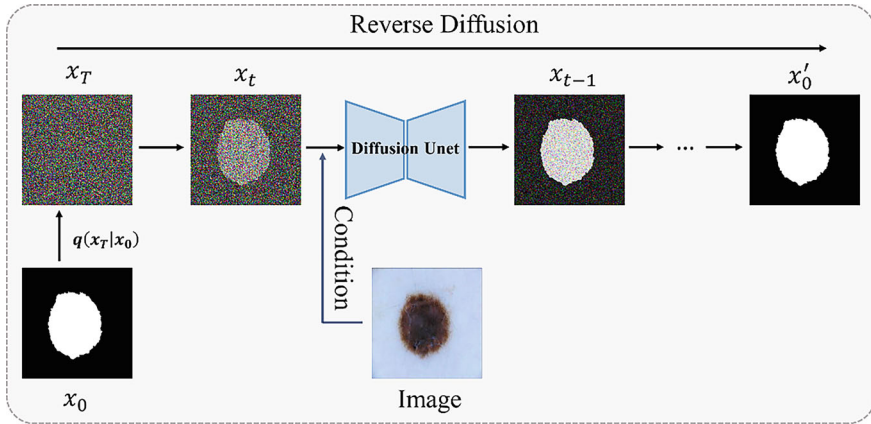


Fig. 2 Overall architecture of the conditional diffusion model for medical image segmentation

where x_t denotes the diffused state at time step t , x_{t-1} is the state at the previous time step, I is the identity matrix, and β_t is a predefined noise injection coefficient that determines the noise magnitude in the transition distribution from x_{t-1} to x_t . In our method, β_t is selected within the range (0.0001, 0.02). After t steps, the forward process can be expressed as:

$$q(x_t|x_0) = N(x_t; \sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)I) \quad (2)$$

Here, α_t denotes the proportion of the data preserved from noise at a single step (“retention coefficient”), and $\bar{\alpha}_t$ is the cumulative retention coefficient, representing the proportion of data not significantly perturbed by noise across previous steps. Utilizing these hyperparameters together with the Parameter Renormalization Technique, we can directly represent:

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \quad (3)$$

2.2 Reverse Diffusion Process

During the reverse generation process in DDPM [16], the original data x_0 is generated by first sampling a noise vector x_T from the prior distribution $p(x_T)$, and then iteratively computing $x_{T-1}, x_{T-2}, \dots, x_1$ using the learned transition kernels, ultimately obtaining x_0 . The goal is to make the reverse Markov chain approximate the “time-reversed process” of the forward chain as closely as possible by adjusting the model parameters θ , so that the joint distribution of the reverse chain $p_\theta(x_0, x_1, \dots, x_T)$ aligns with the joint distribution of the forward chain $q(x_0, x_1, \dots, x_T)$.

The reverse conditional distribution is approximated using a neural network with learnable parameters:

$$p_{\theta}(x_0) = \int p_{\theta}(x_{0:T}) dx_{1:T} \quad (4)$$

where the reverse transition probability $p_{\theta}(x_{t-1}|x_t)$ is Gaussian, enabling stepwise denoising from noise to the original data. Each sampling step is given by:

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1-\alpha_t}{\sqrt{1-\alpha_t}} \epsilon_{\theta}(x_t, t) \right) + \sigma_t z, \quad z \sim \mathcal{N}(0, I) \quad (5)$$

Here, ϵ_{θ} is a deep network parameterized by θ , which predicts the noise ϵ added to x_t in the forward process; z is independently sampled from a standard normal distribution to ensure stochasticity in each reverse step; and σ_t is a learnable noise standard deviation [16], ensuring that the generation process is not purely deterministic but exhibits diversity.

2.3 Loss Function

The training objective is formulated via denoising score matching, minimizing the mean square error (MSE) between predicted and actual noise:

$$L(\theta) = \mathbb{E}_{x_0, \epsilon, t} [\lambda(t) \|\epsilon - \epsilon_{\theta}(x_t, t)\|^2] \quad (6)$$

where $t \sim U_{[1, T]}$ is uniformly sampled from $\{1, 2, 3, \dots, T\}$, $\lambda(t)$ is a positive weighting function, and ϵ_{θ} predicts the forward process noise. The Kullback–Leibler (KL) divergence is introduced to quantify the difference between forward and reverse distributions, and θ is optimized to minimize this divergence. For simplicity and improved sample quality, a simplified loss function using only the noise prediction error is commonly employed:

$$\mathcal{L}_{\text{simple}}(\theta) = \mathbb{E}_{x_0, \epsilon, t} [\|\epsilon - \epsilon_{\theta}(x_t, t)\|^2] \quad (7)$$

3 Experiments and Results

3.1 Experimental Setup

All experiments in this study were implemented using the PyTorch framework and conducted on two NVIDIA RTX A6000 GPUs. To ensure fairness and reproducibility, the model configurations and training protocols strictly followed those

specified in the original papers of the comparative methods, thereby avoiding potential biases caused by implementation differences. Additionally, all experiments were performed under the same hardware and software environment to guarantee the reliability of comparative results. We evaluated the methods on three representative publicly available medical image segmentation datasets:

REFUGE-2 [17]: A fundus image dataset containing 1,600 color retinal images with expert annotations of the optic disc and cup. In this study, the optic cup segmentation subset was selected to evaluate the performance of diffusion models in ophthalmic image segmentation.

ISIC2018 [18]: A dermoscopy image dataset released by ISIC, consisting of 2,594 images with pixel-level masks, primarily used for skin lesion (especially melanoma) segmentation tasks.

Kvasir-SEG [19]: Comprising 1,000 colonoscopy images with accurate pixel-level polyp segmentation annotations, covering diverse polyp morphologies and complex backgrounds, widely used for colorectal lesion detection research.

3.2 *Evaluation Metrics*

To provide a detailed and rigorous assessment of segmentation performance, multiple commonly used evaluation metrics were employed, including Intersection over Union (IoU), Dice Similarity Coefficient (DSC), Positive Predictive Value (PPV), Sensitivity (SEN), and 95% Hausdorff Distance (HD95) [20]. Among these, IoU and Dice measure the degree of overlap between predictions and ground truth, PPV evaluates prediction precision, SEN assesses the model's recall capability, and HD95 reflects the accuracy of boundary delineation. Collectively, these metrics offer a comprehensive evaluation of segmentation results in terms of regional consistency, boundary matching, and classification performance.

3.3 *Comparative Models*

For comparative experiments, we selected several representative diffusion-based segmentation methods, including SegDiff [21], MedSegDiff [12], MedSegDiff-V2 [13], EnsemDiff [14], EIDiffSeg [22], and DiffAtlas [23]. These methods differ in design philosophy and model architecture, representing various directions of diffusion model development in medical image segmentation.

SegDiff, as one of the earliest diffusion-based segmentation frameworks, laid the foundation for subsequent methods. MedSegDiff and its upgraded version MedSegDiff-V2 incorporate conditional encoding and Transformer architectures, effectively improving segmentation stability and accuracy. EnsemDiff leverages

diversified sampling strategies and integrates results to enhance model robustness. EIDiffSeg combines structural priors with the diffusion process, further enhancing the delineation of complex anatomical structures. DiffAtlas is a diffusion-based generative framework that jointly models images and segmentation masks during training, realizing “generative atlas” segmentation. These methods were selected to comprehensively assess the adaptability and advantages of diffusion models across different medical imaging segmentation tasks.

3.4 Experimental Results

To systematically evaluate the effectiveness and robustness of all comparative methods in cross-modal medical image segmentation, experiments were conducted on the REFUGE-2, ISIC2018, and Kvasir-SEG datasets. Fundus, skin lesion, and gastrointestinal images were employed as different segmentation scenarios, comprehensively assessing the adaptability of various models across diverse clinical applications. The results are shown in Tables 1, 2 and 3, respectively.

On the REFUGE-2 dataset, DiffAtlas and EIDiffSeg demonstrated superior performance across multiple metrics. DiffAtlas achieved the best SEN and HD95, while EIDiffSeg slightly outperformed DiffAtlas in Dice, indicating advantages in delineating small structures and complex anatomical boundaries. In contrast, EnsemDiff achieved the highest PPV but exhibited slightly lower overall stability.

Table 1 Segmentation results on the REFUGE-2 dataset

Methods	IoU	Dice	PPV	SEN	HD95
SegDiff [21]	0.7569	0.8587	0.8647	0.8451	10.20
MedSegDiff [12]	0.7625	0.8602	0.8831	0.8573	9.45
MedSegDiff-V2 [13]	0.7686	0.8646	0.8779	0.8692	9.27
EnsemDiff [14]	0.7768	0.8702	0.8929	0.8353	8.02
EIDiffSeg [22]	0.7913	0.8796	0.8849	0.8920	7.51
DiffAtlas [23]	0.7930	0.8781	0.8903	0.8939	7.08

Table 2 Segmentation results on the ISIC2018 dataset

Methods	IoU	Dice	PPV	SEN	HD95
SegDiff [21]	0.7831	0.8546	0.8985	0.8591	16.45
MedSegDiff [12]	0.7972	0.8692	0.9168	0.8674	15.90
MedSegDiff-V2 [13]	0.8038	0.8732	0.9207	0.8680	15.29
EnsemDiff [14]	0.8250	0.8920	0.9402	0.8748	12.61
EIDiffSeg [22]	0.8324	0.8975	0.9284	0.8859	12.11
DiffAtlas [23]	0.8335	0.8993	0.9321	0.8894	11.87

Table 3 Segmentation results on the Kvasir-SEG dataset

Methods	IoU	Dice	PPV	SEN	HD95
SegDiff [21]	0.7628	0.8319	0.8466	0.8350	27.67
MedSegDiff [12]	0.7730	0.8424	0.8561	0.8431	26.91
MedSegDiff-V2 [13]	0.7832	0.8511	0.8705	0.8478	25.11
EnsemDiff [14]	0.7893	0.8987	0.8841	0.8509	24.93
EIDiffSeg [22]	0.8020	0.8711	0.9232	0.8613	22.26
DiffAtlas [23]	0.8105	0.8758	0.9253	0.8645	21.88

On the ISIC2018 dataset, DiffAtlas again attained the highest IoU and Dice, with an HD95 significantly lower than other methods, demonstrating greater precision in skin lesion boundary reconstruction and region overlap. Although EnsemDiff excelled in PPV, its SEN was slightly lower than DiffAtlas, suggesting a tendency toward reducing false positives at the expense of marginally increased false negatives.

On the Kvasir-SEG dataset, DiffAtlas showed the most robust performance, achieving the best scores in all the evaluation metrics, demonstrating robustness under complex backgrounds and diverse polyp morphologies. EIDiffSeg ranked closely behind, whereas SegDiff and the MedSegDiff series showed relatively weaker overall performance.

4 Conclusion

This study addresses the limitations of existing diffusion-based medical image segmentation methods, which often focus on single diseases and specific datasets. Cross-task experiments were conducted on REFUGE-2, ISIC2018, and Kvasir-SEG to evaluate model performance. DiffAtlas consistently achieved the best overall results, leading in region-overlap metrics (IoU, Dice) and demonstrating strong stability and generalization in fine-grained lesion recognition and boundary delineation (SEN, HD95). EIDiffSeg showed comparable Dice and SEN, indicating potential in modeling complex anatomical structures. EnsemDiff excelled in PPV, highlighting the benefits of ensemble strategies in reducing false positives, though with lower overall robustness. SegDiff and MedSegDiff series performed less effectively, reflecting limitations in generalization and boundary-detail handling.

Overall, the findings validate the effectiveness and robustness of diffusion-based segmentation methods in multi-modal medical imaging and underscore the advantages of generative frameworks that jointly model images and masks, particularly for limited-sample, cross-domain, and complex structural segmentation tasks.

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